

National University Management System

1. Project Title and Description

Project Title

‘National University Management System’

Description

The National University Management System is a database-driven application designed to manage university operations efficiently. It stores and manages information related to departments, teachers, students, courses, classrooms, enrollments, attendance, examination results, fee records, and user accounts. The system helps administrators, teachers, and students perform academic and administrative tasks through a centralized database.

2. Scenario Explanation

A university manages thousands of students, teachers, courses, and academic records every semester. Handling this information manually can lead to errors, duplication, and inefficiency.

The National University Management System provides a centralized database where:

- Departments manage academic programs.
- Teachers teach courses and maintain attendance.
- Students enroll in courses and view results.
- Administrators manage fees, classrooms, and users.
- Academic records are stored securely and can be retrieved quickly.
- This system ensures data consistency, accuracy, and efficient management of university operations.

3. List of All 10 Features

Feature 1: Create New Table

Purpose: Create a new database table such as Hostel.

What it does:

- Creates a new table structure.
- Defines columns and constraints.
- Stores additional university data.

Feature 2: Insert Data

Purpose: Add new student records.

What it does:

- Stores student information.
- Adds new entries into the database.
- Supports student registration.

Feature 3: View All Students

Purpose: Display all student records.

What it does:

- Retrieves all student information.
- Allows administrators to view records.
- Helps in monitoring enrolled students.

Feature 4: Search Student

Purpose: Find a specific student.

What it does:

- Searches using Registration Number.
- Displays matching student details.
- Helps locate student records quickly.

Feature 5: Student-Department Join

Purpose: Display students with their department names.

What it does:

- Combines data from Students and Departments tables.
- Shows department information alongside student details.
- Demonstrates relational database functionality.

Feature 6: Update Student Information

Purpose: Modify existing student records.

What it does:

- Updates semester information.

- Corrects student data.
- Keeps records current.

Feature 7: Delete Student Record

Purpose: Remove unnecessary records.

What it does:

- Deletes inactive students.
- Maintains database cleanliness.
- Removes outdated information.

Feature 8: Student Count by Department

Purpose: Generate departmental statistics.

What it does:

- Counts students in each department.
- Produces summary reports.
- Helps administration analyze enrollment.

Feature 9: Modify Table Structure

Purpose: Add new attributes.

What it does:

- Adds CGPA column to Students table.
- Enhances database functionality.
- Supports future requirements.

Feature 10: User Access Control

Purpose: Manage database permissions.

What it does:

- Grants access rights to users.
- Controls database security.
- Restricts unauthorized operations.

4. Database Schema Overview

Tables

- i. Departments**
 - Stores department information.
 - Primary Key: dept_id
- ii. Teachers**
 - Stores teacher information.
 - Foreign Key: dept_id
- iii. Students**
 - Stores student records.
 - Foreign Key: dept_id
- iv. Courses**
 - Stores course details.
 - Foreign Keys: dept_id, teacher_id
- v. Classrooms**
 - Stores classroom information.
- vi. Enrollments**
 - Manages student-course registrations.
 - Foreign Keys: student_id, course_id
- vii. Attendance**
 - Stores attendance records.
 - Foreign Keys: student_id, course_id
- viii. Results**
 - Stores examination results.
 - Foreign Keys: student_id, course_id
- ix. Fee_Records**
 - Stores fee payment information.
 - Foreign Key: student_id
- x. Users**
 - Stores login and access control information.

Relationships

- One Department → Many Teachers
- One Department → Many Students
- One Department → Many Courses
- One Teacher → Many Courses
- Many Students ↔ Many Courses (through Enrollments)
- One Student → Many Attendance Records

- One Student → Many Results
- One Student → Many Fee Records

5. Explanation of SQL Commands Used

i. ALTER TABLE

Meaning

ALTER TABLE is used to modify the structure of an existing table.

Purpose

It allows adding, modifying, or deleting columns and constraints.

Where Used

- Adding a CGPA column to the Students table
- Adding new attributes to Teachers or Courses tables

ii. INSERT INTO

Meaning

INSERT INTO is used to add new records to a table.

Purpose

It stores new data in the database.

Where Used

- Students
- Teachers
- Departments
- Courses
- Attendance
- Results
- Fee Records

iii. SELECT

Meaning

SELECT is used to retrieve data from a database table.

Purpose

It displays stored information for users and administrators.

Where Used

- Viewing student records
- Viewing teacher records
- Viewing course information
- Viewing fee records

iv. WHERE

Meaning

WHERE is used to apply conditions and filter records.

Purpose

It retrieves only the data that meets specific criteria.

Where Used

- Searching students by Registration Number
- Finding unpaid fee records
- Finding a specific teacher or course

v. UPDATE

Meaning

UPDATE is used to modify existing records in a table.

Purpose

It keeps database information accurate and up to date.

Where Used

- Updating student semester
- Updating fee status
- Updating teacher designation

vi. DELETE

Meaning

DELETE is used to remove records from a table.

Purpose

It removes outdated, duplicate, or unnecessary data.

Where Used

- Removing inactive students
- Deleting duplicate records
- Removing old attendance entries

vii. JOIN

Meaning

JOIN is used to combine data from two or more related tables.

Purpose

It allows retrieval of related information stored in different tables.

Where Used

- Students and Departments
- Courses and Teachers
- Results and Students
- Enrollments and Courses

viii. GROUP BY and COUNT

Meaning

GROUP BY groups similar records together, while COUNT calculates the number of records in each group.

Purpose

It is used for reporting, analysis, and statistical summaries.

Where Used

- Counting students in each department
- Counting courses in each department
- Attendance statistics

ix. GRANT

Meaning

GRANT is used to provide permissions to database users.

Purpose

It controls access and improves database security.

Where Used

- Providing teacher access
- Providing administrator access
- Managing user privileges

x. CREATE TABLE

Meaning

CREATE TABLE is used to create a new table in the database.

Purpose

It defines the structure of a table, including columns, data types, and constraints.

Where Used

- Departments
- Teachers
- Students
- Courses
- Classrooms
- Enrollments
- Attendance
- Results
- Fee_Records
- Users

